gemini-conversation

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User

how to calculate doppler from consecutive radar pulse returns and explain what is doppler frequency

Gemini

What is Doppler Frequency?

At its core, **Doppler frequency** (or Doppler shift) is the change in frequency of a wave in relation to an observer who is moving relative to the wave source.

In radar systems, the radar transmits a radio frequency (RF) signal at a stable carrier frequency f_c. When this signal hits a target and bounces back, the frequency of the returned signal shifts if the target is moving relative to the radar:

- **Target moving toward the radar:** The wave crests are compressed, resulting in a higher received frequency (positive Doppler shift).
- **Target moving away from the radar:** The wave crests are stretched out, resulting in a lower received frequency (negative Doppler shift).

The relationship between the Doppler frequency shift f_d and the target's radial velocity v_r (the velocity component directly along the radar's line of sight) is given by:

$$f_d = \frac{2v_r}{\lambda} = \frac{2v_r f_c}{c}$$

Where:

- v_r is the radial velocity of the target.
- λ is the wavelength of the transmitted carrier signal.
- c is the speed of light ($\approx 3 \times 10^8 \text{ m/s}$).
- The factor of **2** accounts for the two-way round-trip path of the radar signal (out to the target and back).

How to Calculate Doppler from Consecutive Pulse Returns

While you could try to measure a tiny shift in the carrier frequency f_c directly, at microwave frequencies (e.g., 10 GHz), a target moving at 100 m/s only causes a shift of about 6.7 kHz. Measuring such a minute fractional change directly on a single pulse is extremely difficult and highly susceptible to noise.

Instead, radars calculate Doppler by measuring the **pulse-to-pulse phase change** over a sequence of consecutive pulses. This sequence of pulses is known as a **Pulse Train** or **Coherent Processing Interval (CPI)**.

Step 1: Mix to Baseband (I/Q Demodulation)

When a pulse returns, the receiver mixes it with a local oscillator to strip away the carrier frequency f_c. This yields a complex baseband signal representing the **In-phase (I)** and **Quadrature (Q)** components:

$$y(t) = I(t) + jQ(t) = Ae^{j\phi(t)}$$

For a single target at range R, the phase ϕ of the return signal is directly proportional to the round-trip distance the wave traveled:

$$\phi = 2\pi \cdot \frac{\text{Round-trip Distance}}{\text{Wavelength}} = 2\pi\frac{2R}{\lambda} = \frac{4\pi R}{\lambda}$$

Step 2: Sample in "Fast-Time" and "Slow-Time"

To process multiple pulses, we organize the received data into a 2D matrix:

- **Fast-Time (Range):** Sampling within a single pulse return to determine target range.
- **Slow-Time (Pulses):** Sampling the same range bin across consecutive pulse returns. If the Pulse Repetition Interval is T (where $T = \text{PRI}$), then pulse m occurs at slow-time $t_m = m \cdot T$.

If a target is moving with a constant radial velocity v_r, its range changes over time: $R(t) = R_0 - v_r t$.

Substituting this into the phase equation for pulse m:

$$\phi(m) = \frac{4\pi(R_0 - v_r mT)}{\lambda} = \frac{4\pi R_0}{\lambda} - \left(\frac{4\pi v_r T}{\lambda}\right) m$$

Notice that the phase changes linearly from pulse to pulse! The phase difference $\Delta\phi$ between two consecutive pulses (m and $m-1$) is:

$$\Delta\phi = \phi(m) - \phi(m-1) = -\frac{4\pi v_r T}{\lambda}$$

Using the Doppler relation $f_d = \frac{2v_r}{\lambda}$, we can rewrite this as:

$$\Delta\phi = -2\pi f_d T$$

Step 3: Extract the Frequency via the FFT

Because we have a series of discrete complex samples over slow-time m for a specific range bin:

$$x[m] = Ae^{j(\phi_0 - 2\pi f_d T m)}$$

Calculating the Doppler frequency is equivalent to finding the dominant frequency of this discrete complex exponential sequence.

We do this by performing a **Discrete Fourier Transform (DFT)**—typically implemented as a **Fast Fourier Transform (FFT)**—across the slow-time dimension (the consecutive pulses) for that range bin. This is often called a **Doppler FFT** or **pulse-Doppler**

processing.

```
Consecutive Pulses (Slow-Time)
   Pulse 1      Pulse 2      Pulse 3         Pulse M
[ I_1 + jQ_1 ,  I_2 + jQ_2 ,  I_3 + jQ_3 , ... , I_M + jQ_M ]
     │
     ▼
┌──────────┐
│   FFT    │  (Spectral Analysis)
└────┬─────┘
     ▼
Doppler Spectrum Bin Peak ──► Identifies the Doppler Frequency (fd)
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The peak index k in the resulting FFT spectrum corresponds to the digital frequency, which maps to the physical Doppler frequency:

$$f_d = \frac{k}{M \cdot T} = k \cdot \frac{\mathrm{PRF}}{M}$$

Where:

- M is the number of pulses processed in the FFT.
- $\mathrm{PRF} = 1/T$ is the Pulse Repetition Frequency.
- k is the bin index of the peak in the FFT output.

Crucial Constraints: Ambiguity and Resolution

When calculating Doppler from pulse returns, you must manage two fundamental physical limits:

Doppler Resolution (Δf_d)	$\Delta f_d = \frac{1}{M \cdot T} = \frac{1}{\mathrm{CPI}}$	How close two targets can be in velocity and still be resolved as separate. To get better resolution, you must observe the target for a longer time (CPI).

Maximum Unambiguous Doppler ($f_{d,\max}$)	$f_{d,\max} = \pm\frac{\mathrm{PRF}}{2}$	Because we sample at interval T, the Nyquist sampling theorem limits the maximum detectable phase shift to $\pm\pi$ radians per pulse. Any shift beyond this "wraps around" (aliases).